

ORNL/TM-8107
Dist. Category UC-20

ORNL/TM--8107

DE83 005160

Contract No. W-7405-eng-26

FUSION ENERGY DIVISION

THE ND6600 COMPUTER IN FUSION-ENERGY RESEARCH

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Date Published - December 1982

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Oak Ridge, Tennessee 37830
operated by
UNION CARBIDE CORPORATION
for the
DEPARTMENT OF ENERGY

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ABSTRACT

The ND6600, a computer-based multichannel analyzer with eight ADCs, is used to acquire x-ray data. This manual introduces a user to the Nuclear Data system and contains the information necessary for the user to acquire, display, and record data. The manual also guides the programmer in the hardware and software maintenance of the system.

1. INTRODUCTION

The ND6600 in the Fusion Energy Division (FED) is used to acquire x-ray data from the ELMO Bumpy Torus (EBT). The EBT is a steady-state toroidal device which uses a magnetic field to contain a hot plasma. The plasma is heated primarily by microwaves. The x-ray data taken with the ND6600 can be analyzed to yield the temperature and density of the plasma as a function of machine conditions (microwave power, frequency, etc.). The purpose of the EBT research project is to develop a viable and economically feasible fusion energy device.

This manual describes the ND6600 as used in FED. In particular, it summarizes and reorganizes the information given in the Nuclear Data (ND) manuals, emphasizing those aspects used in our application and omitting those not used. In addition, this manual contains information not given in the Nuclear Data manuals as well as a discussion of those features or software created specifically for our system.

This manual has been organized so that a user who wishes to start by taking data can turn directly to the section on data handling and begin using the system. It is recommended that the sections on hardware and software be read before reading the remaining sections, which may be read in any order.

2. HARDWARE

The ND6600 is a computer-based multichannel analyzer. The basic system components include a system central processing unit (CPU), system memory, a display and acquisition subsystem (DAS), a COMBUS (defined below), printed circuit boards (listed below), and various peripherals.

The peripherals consist of a dual hard disk and a dual floppy disk unit. The hard disk system consists of one fixed and one removable platter. The disks are 24-sector, 2.5-megabyte disks (2315 or IBM 5440 disk cartridge). The floppy disks have a storage capacity of 256,256 bytes and are interchangeable with the diskettes used in the IBM 3740 (Shugert SA100, Scotch 704, or equivalent).

The solid-state random access memory (64,000 16-bit words) is contained on two printed circuit boards.

The COMBUS is the vehicle through which data communication between all system components is accomplished and through which power is distributed. The COMBUS is a nondedicated, high speed, decentralized, synchronous bus which permits direct, bidirectional data communications between the components of the ND6600 system. It consists of a printed circuit board, containing 100 bus lines and twenty-three 100-pin connectors for insertion of plug-in system component boards.

The system CPU directs the task performance of the device microprocessor controllers within the ND6600 system. It consists of a Digital Equipment Corporation (DEC) LSI-11 microcomputer with Firmware Extended and Floating Point Instruction Sets attached to a printed circuit board containing the System Memory Management and Interface circuitry. The LSI-11 is a 16-bit microcomputer with the speed and instructions of a minicomputer. Its extensive instruction set of more than 400 instructions is equivalent to the PDP-11/35 or 40 computer system. The only departure from the standard PDP-11/35 or 40 CPU is the addition of two new instructions, used explicitly to access the processor status word.

The DAS acquires data directly into system memory and provides alphanumeric and graphic display. The DAS has its own LSI-11 and runs under control of a program stored in this LSI-11. Thus, display and acquisition are done independently of processing and I/O.

The boards installed in the COMBUS are listed below.

MMI + LSI-11	Memory management interface
DIB + LSI-11	DAS interface board
HDB + MIP	Hard disk interface and microprocessor board
FDB + MIP	Floppy disk interface and microprocessor board
IOB	Input/output board — provides ports for paper tape, punch, and line printer and two ports for RS-232, 20-MA current, or optical coupling
DKI	DAS keyboard control
GPD	General purpose display
DAD	DAS acquisition and display — has ports for 4 ADCs
SCP + MIX	Serial channel processor for communications
SKM, FPM	Solid-state memory

3. SOFTWARE - THE OPERATING SYSTEM

The multitask operating system is called MIDASD+. The maximum number of users is 256, and a single MIDAS user can execute 1-3 programs concurrently. The operating system is terminal oriented and interacts with the user in a conversational mode. As an alternative to the interactive mode of operating the system, the user may create a job stream by putting MIDAS commands in a file and having them automatically executed by the system. In this section we will give

examples of disk facilities, logical units, and ownership and protection of files.

In the following, a command that the user types is indicated in all capital letters and must be followed by a carriage return (not indicated). The system's response is in small letters. The symbol > is the system's prompt and means that the user may enter commands. The symbol # means the system is waiting for input for a particular program.

3.1 DISK FACILITIES

Before anything can be written on disk, a file must be created. The file can be directoried or nondirectoried. If it is nondirectoried it can contain only one program or piece of data. If it is directoried it can contain more than one piece or element. If the file is called A and the element B, the piece of data is said to have name A.B.

A hard disk has 9743 sectors. A sector has 256 bytes. A file may be created by using the system command FILE:

```
FILE  A,DK1,200,400,D
```

This command will create a file named A on hard disk #1, at starting sector 200. It will be 400 sectors long and will be directoried. Writing to this file is discussed in the sections on data handling and Fortran programming. DK0 refers to the fixed disk upon which the operating system resides. DK1 refers to the removable hard disk. Normally, files are created to fill the entire disk and are rarely changed. Elements are deleted as necessary using the command


```
DEL A,B,DK1
```

After an element is deleted, the file may be packed, thus eliminating the deleted element:

```
PACK A,DK1
```

Occasionally, one may wish to delete all the elements in a file with one command. This can be done by typing

```
DEL A*,DK1
```

Then, one must recreate the file using the FILE command. Before doing this, it is advisable to find the starting sector and length of the file. This may be done by typing

```
DIR A
```

3.2 LOGICAL UNITS

In order to write to a file, teletype, line printer, or other device, one assigns the device a logical unit number. This is done with the command

```
DEF 2,A.B or DEF 2,TTY
```

The DAS terminal is called TTY. The DECwriter III attached to port 1 on the IOB is called TT1.

To assign a logical unit number in a Fortran program, use the subroutine DEFY. CALL DEFY(LUN,NAME) where LUN is the logical unit number and NAME is a logical*1 NAME(16) array.

3.3 PASSWORDS, PROTECTION, AND OWNERSHIP

This section will discuss the assignment of passwords and the meaning of protection and ownership. For more detailed information see the ND manuals.

3.3.1 Passwords

To sign on to use the ND6600, type

HEL N

where N is the user number. To sign off, type

BYE N

To find which user is signed on to the ND, type WHO.

When the ND6600 is shipped from the factory, the following passwords and priorities are in effect:

User Number	Password	Priority
0	MIDAS	1
1	USER1	2
2	USER2	2

The priorities range from 1 through 15, with 1 the highest. User numbers range from 1 to 255. To assign a new user number, sign on as the control operator and type

```
ASSIGN N,PR,PW,SF
```

where

N = User number

PR = Priority

PW = Password

SF = User's auxiliary command table

The standard user's table is SYSFIL.USEELM. For example, to assign user number 20 a priority of 2, a password GUESS, and the standard user's table, type

```
ASSIGN 20,2,GUESS,SYSFIL.USEELM
```

3.3.2 Protection and Ownership

An element of a file is owned by the person who copies or creates it. If it is write protected, the owner may write to it, but another user will receive an error message:

?Err 35 Attempt to output to read only file

Any time a user copies a file.element, it is automatically write protected. To determine the ownership and protection of a file.element, type

FDI FNAME.ELM

where FNAME.ELM is the name of the file.element. Note that the entire file may be write protected, but it is the status of the individual elements that counts. For example, the file TEST may be write protected, but the element TEST.A001 may not be. Also, the owner 255 is considered to be a universal owner. Anyone may write in a file owned by #255.

4. DATA HANDLING

4.1 INITIALIZATION

To use the ND6600, first sign on. To sign on, enter the user number and password (see above). If the user number is 2 and the password is user2, enter

HEL 2

USER2

To check the date and time, enter

DAT

TIM

To set the date and time for October 1, 1981, at 1:30:33, enter

DAT 1 OCT 81

TIM 13:30:33

Before taking data, assign the selected ADC with selected number of groups, number of channels per group (1-8192), number of bits per channel (1-30), and preset live time:

AADC 1,4,256,30,300

This assigns ADC number 1 with four groups of 256 channels each, with 30 bits (max count = 2^{*30}) per channel for 300 seconds. Alternatively, if one wished to assign six ADCs one could use a job stream (see Sect. 5). For example, the job stream A4G16 has been written to assign ADCs 1 through 6 with the parameters given above.

Thus, one would type

JOB AUTO.A4G16

To erase group 1 of all assigned ADCs, hit the button marked ERA on the DAS terminal. To erase group G of ADC A, type

ERA A/G.

4.2 ACQUISITION

To acquire data in group 1 of all assigned ADCs, type

ADON

Data will accumulate until the preset time elapses. A red light will appear on the ADC indicating it is in acquisition mode. To stop, type

AOFF

If the preset time has not elapsed, the next time one types ADON, the timer will continue at the time it left off.

There are two ways to acquire data in group G of ADC A. One is to type

ADON A/G

To stop, type

AOFF A/G

The other is to press the key marked ACQ and then press the key marked INIT. This will acquire data in the ADC and group marked in the display (see Sect. 4.3, Display, below). To stop, press ACQ and INIT again.

4.3 DISPLAY

This section will discuss how to read and display one or two spectra stored on disk and how to expand a region of the spectrum. For reference, a diagram of the keypad and the display are shown in Figs. 1 and 2.

4.3.1 How to Display One Spectrum

To display the spectrum DATA.A001, which is recorded on disk, first assign an ADC:

AADC 1,4,256,30,300

Next, define logical unit 12:

DEFINE 12 DATA.A001

Then, run the program to read the spectrum:

RUN APS.READ

The program will pause for you to enter the ADC number and the group number. A response of

1/1

will read the spectrum into group 1 of ADC 1. The spectrum should appear on the screen. If you can't see it, try the log scale by pressing the key marked LOG on the keypad.

The display area of the screen can be made smaller or larger by pressing the toggle key labelled +/- and then the key labelled PAGE. Pressing the key marked GROUP will change the group number being displayed by one. Also, if more than one ADC is assigned, pressing the key labelled ADC will change the ADC number by one. The current value of the ADC and group being displayed is recorded to the left of the graph of data.

4.3.2 How to Display Two Spectra

Two spectra may be displayed simultaneously -- one blinking and the other stationary. The stationary one is called the main spectrum and the blinking one is called the overlap spectrum. The following steps should be followed:

- (1) Assign an ADC:

AADC 1 4 256 30 300

- (2) Set the scale to log:

Press the key marked LOG on the keypad

- (3) Erase the memory

ERA

- (4) Assign a second ADC:

AADC 2 4 256 30 300

- (5) Press the key marked main/olap on the keypad -- note that the space labelled FKC in the upper left of the screen displays SO+:

Press MAIN/OLAP

- (6) Press the key marked INIT:

Press INIT

- (7) Press the keys marked +/- and ADC until the space labelled ADC in the left of the display reads ADC D1-1 D1-2:

Press +/- and ADC

- (8) Press the keys +/- and GRP until the space labelled GRP reads GRP
1 3:

Press +/- and GRP

- (9) Press the key marked log:

Press LOG

- (10) Type:

ERA 2/3

- (11) To read spectrum DATA.A001 into the main area, type

DEF 12 DATA.A001

R READ

1/1

- (12) To read spectrum DATA.A002 into the overlap area, type

DEF 12 DATA.A002

R READ

2/3

To remove the overlap spectrum, press the keys sptm/regm and main/olap
and +/- until the area FKC reads SO-; then, press INIT.

4.3.3 How to Expand a Region of the Spectrum

Assign ADCs and read in the desired spectrum as described above.

To expand the spectrum in the vertical direction:

- (1) Set the sign in the region labelled FKC with the +/- toggle switch to read +.
- (2) Set the spectrum on linear scale with the toggle switch marked LOG.
- (3) Hold down the key marked $\uparrow\downarrow$ until the spectrum is visible and the desired size.

To shrink the spectrum:

- (1) Press the toggle marked +/- so that the sign in the area labelled FKC is -.
- (2) Press the key marked $\uparrow\downarrow$.

To expand or contract in the horizontal direction, use the key marked $\leftarrow\rightarrow$ instead of the key $\uparrow\downarrow$. To expand or contract in both directions, use the key marked $\leftrightarrow$. If you lose sight of the spectrum, set the scale to log with the LOG key (an L for log will appear inside the spectral box at the top), set the FKC area to read SM- (use the keys marked +/-, SPTM/RGN and MAIN/OLAP), and depress the key marked $\leftrightarrow$.

An expanded region of the spectrum can be displayed at the same time as the entire spectrum. To do this, use the toggle switch marked SPTM/REGN. The FKC area should read RM+. Then, press the key marked INIT. The entire spectrum will appear in a box to the upper right, as shown in Fig. 2. If it is not visible, put it on LOG scale where it may be easier to see. To select a particular region of the spectrum, change + to - with the +/- toggle switch. To shorten the region vertically:

Press the keys to obtain RM- in the FKC area

and then press the key $\begin{smallmatrix} \uparrow\uparrow \\ \downarrow\downarrow \end{smallmatrix}$.

To narrow the region horizontally:

Press the keys to obtain RM- in FKC area

and then press the key labelled E.

To move the region to the right:

Press the keys to obtain RM+ in the FKC area

and then press the key labelled C.

To move the region to the left:

Press the keys to obtain RM- in the FKC area

and then press the key labelled C.

To move the region up and down, use the +/- key with the key B.

4.4 MANIPULATION

It is possible to add, subtract, multiply, or divide spectra using the Nuclear Data program APS.MANIP. The spectra may be stored in memory groups or in files on disk. For example, to subtract the

spectrum in ADC1 group 2 from the spectrum in ADC1 group 1 and leave the result in ADC1 group 3, enter commands as follows:

RUN APS.MANIP

Mov to # 1/3

Results of operand 1 # 1/1

Operator # -

Operand 2 # 1/2

To add two spectra, DATA.A and DATA.B, stored on disk and leave the results on disk in DATA.C:

RUN APS.MANIP

Mov to # DATA.C

Results of operand 1 # DATA.A

Operator # +

Operand 2 # DATA.B

To move the spectrum in ADC1 group 1 to ADC1 group 2, multiply the spectrum by 1:

RUN APS.MANIP

Mov to # 1/2

Results of operand 1 # 1/1

Operator # *

Operand 2 # 1

Further examples and discussion may be found in the manual ND6600 Data Acquisition and Processing System, p. 8-13.

4.5 RECORDING THE DATA

There are two ways to record data: one is to use job streams and the other is to use the ND6600 programs APS.WRITE and APS.READ. The use of job streams is an automatic way of using APS.WRITE and APS.READ. Thus, it is faster and job streams have been written for those cases we use most frequently. We will first describe how to run the job streams (how to write job streams is discussed in the next section); then, we will discuss the use of the program APS.WRITE. The programs APS.READ and APS.PRINT will also be described.

Before recording the data, first set the sequence number for the spectrum names:

SET ,,VAL

If VAL = 0, the first spectrum will be written on disk as DATA.A0001.

To record one 256-channel spectrum in the file DATA, type

JOB AUTO.R #

where # is the ADC number. This will write the data in ADC1 group 1 into spectrum DATA.A0001. To check if it is really there, read it back in. First erase the desired ADC and group:

ERA A/G

Then, define the logical unit number:

DEF 12,DATA.A0001

Run the program to read it:

RUN APS.READ

1/2

will read the spectrum into ADC1 group 2.

To record six 256-channel spectra taken with ADCs 1 through 6 and stored in groups 1, type

JOB AUTO.R1T6

To record spectra with background subtracted for six ADCs:

(1) Acquire backgrounds in groups 1 of the six ADCs

(2) Type

JOB AUTO.MOVIT

to move the background from group 1 to group 4

(3) Acquire data in group 1

(4) Type

JOB AUTO.RECIT

This job stream RECIT does three things: (1) it moves the data from groups 1 to groups 3; (2) it subtracts the background and stores the result in groups 2; and (3) it records results only in DATA.S # # # #, where # # # # is the current number of the spectrum. Therefore, if the test word was to be set at 6, the first spectrum recorded will be DATA.S0007 and the last will be DATA.S0012. To check on this, type

DIR DATA

This will give the directory or list of elements recorded in DATA. The user may also print the contents of a spectrum on the terminal with the program PRINT:

RUN APS.PRINT

Enter data source and locations: # DATA.S0007,1,40

This will print the contents of channels 1 through 40 of the spectrum DATA.S0007.

The Nuclear Data programs READ, WRITE, and PRINT all use the first two sectors of a disk file/element for header information and the rest of the element for spectral data. To use the program WRITE it is necessary to define two logical unit numbers, 8 and 12. The one to which the data will be written is always 12:

DEF 12,DATA.A0001

The one which contains header information used by Nuclear Data is 8:

DEF 8,EFF.MAIN1

Then,

RUN APS.WRITE

Thus, the program WRITE reads system parameters from the header of EFF.MAIN1 and puts them in the header of DATA.A0001. It also reads parameters from memory (ADC #, time elapsed, etc.) and places them in the header of A0001. Then, it reads the spectral data from memory and places it in the spectrum DATA.A0001. Note that when we bought the system there was already a spectrum recorded on it which was used for the original header information. (For more information see the part

disk file format in the section on Fortran programming.) Also note that before we can write to DATA.A0001 it must exist. There are two ways to create it. One is to copy an existing spectrum and then overwrite it; thus,

```
COP DATA.A0099,DATA.A0001
```

creates DATA.A0001. The second is to create the file.element:

```
RES DATA.A0001,BLK=6
```

for a 256-channel spectrum. If the number of channels differs from 256, calculate the number of blocks using the following formula:

$$\text{BLK} = 2 + \text{\#channels}/64.$$

5. JOB STREAMS

In this section we will discuss job streams (including looping, chaining, comments, and internal file name generation) and give examples of each.

A job stream is entered with the editor (see section on Fortran programming) and then copied into the file AUTO. Thus, if one created a job stream to subtract the spectrum in ADC1 group 2 from the spectrum in ADC1 group 1 and to place the result in ADC1 group 3, it would contain the following:

```
RUN APS.MANIP
```

23

1/3

1/1

-

1/2

RUN APS.COM

Subtraction ADC1 G3 = ADC1 G1 - ADC1 G2

ENDJOB

Thus, one enters the input for the program one wishes to run. The program APS.COM is a job stream program which will output the next line on the terminal. ENDJOB indicates the end of the job stream. To execute the job stream, type

JOB AUTO.A

Error messages are not printed on the terminal. To see error messages, use the command

JOB AUTO.A,.,TTY.

This command also prints each line of the job stream as it is executed.

Looping may be accomplished by using statement labels and GOTO statements, i.e.,

```
K)RUN APS.SEQ
```

```
.
```

```
.
```

```
.
```

```
GOTO K-
```

The - indicates that the label K precedes the GOTO statement. If one wished to execute this loop ten times, one would enter

```
SET,,10
```

```
K)RUN APS.SEQ
```

```
.
```

```
.
```

```
.
```

```
GOTO K-SOB
```

SET sets a test word to 10; SOB subtracts and branches until the test word equals 0.

Job streams can be chained or run successively by placing the command

```
CHAIN AUTO.B
```

immediately before the ENDJOB. Then, instead of exiting, the system goes to job AUTO.B.

To generate element names in the file DATA, use the program APS.SEQ. First use SET to set the original sequence number:

```
SET, ,5
RUN APS.SEQ
DEF 12,DATA.A****
```

This will define logical unit 12 to have the name DATA.A0005. The command

```
SET ADA 1
```

will increment the test word used for the sequence number by one. One can also automatically decrement the test word and branch to label A by using the command

```
GOTO A-SOB.
```

6. PROGRAM GENERATION

6.1 THE EDITOR

The purpose of the ND6600 text editor is to create and modify ASCII files. To use the text editor on the ND6600, first set up an input and an output file:

```
DEF 1,SOURCE.TEST
```

DEF 2,SCR1

If the user already has something in SOURCE.TEST he can read it in and modify it. Otherwise he can simply write information to SCR1. Thus,

```
RUN APS.EDITOR
ND-66    text editor
using lu-1 for input, lu-2 for output

unit      file      element
01        source    test
02        scr1
#
```

To read in the file SOURCE.TEST, type R. To insert code, type I:

```
I
WRITE(7,100)
100FORMAT (' THIS IS A TEST ')
END
]

EX
```

The] stands for escape and takes the user from edit to command mode. EX causes an exit from the program. The program is now in SCR1. To save it, enter

COP SCR1,SOURCE.TEST

This assumes that the file SOURCE exists, but the element TEST does not. If SOURCE.TEST exists, use the command

REP SCR1,SOURCE.TEST.

For those familiar with TECO, the following conversion table containing some of the commands used in both editors may be useful:

TECO symbol	ND symbol	Definition
L	A	advance a line
J	B	go to beginning
ZJ	/J	go to end
FSstr]sta]	Gstr]=Csta	change str to sta
I	I	insert
]]	]	go from edit to command mode
EX]]	EX <u>CR</u>	exit (<u>CR</u> = carriage return)
control C	EN	exit without change
T	V	list or verify
K	K	kill a line

For further details and commands, see the ND6600 manual Data Acquisition and Processing, p. 6-3.

6.2 GENERATING A PROGRAM

Creating and running a Fortran program consists basically of six steps:

- (1) Define necessary logical units.
- (2) Enter the program with the editor.
- (3) Compile the program, producing the object code.
- (4) Set up a link file.
- (5) Link the object code to produce the executable code.
- (6) Run the program.

6.2.1 Define the Logical Units

```
DEF 1 SOURCE.TEST
DEF 2 SCR1
DEF 3 SCR2
DEF 5 TTY
DEF 6 TTY
DEF 7 TTY
DEF 8 LINK.TEST
DEF 11 SCR3
```

The editor uses logical unit 1 for input and logical unit 2 for output. Logical unit 2 is also the input for the compiler. Logical unit 3 is defined to be the output of the compiler or object file. Logical units 5, 6, and 7 are defined to be the teletype. Logical unit 8 is the link file, containing among other things the list of object

files to be linked. The linker writes the executable code on logical unit 11. To list the logical units, type

LUP

6.2.2 Enter the Program With the Editor

Assume the program consists of a main routine and a subroutine. Enter and compile the main routine first:

```
R EDITOR
I
A=2
B=3
CALL ADDIT(A,B,C)
WRITE (7,100) C
100  FORMAT(' THE ANSWER IS ',F10.3)
END
]
EX
```

The source code is now in SCR1, assigned to LU-2.

6.2.3 Compile the Program

Running the Midas/Fortran compiler takes the source code from LU-2 and produces the object code on LU-3:

```
R FORTRAN
```

```
#CR(no switch options)
```

This produces a printout on the file specified for LU-6 with error messages (if any) and a symbol table of the source code. Midas/Fortran error codes and definitions are given in the manual Program Generation, p. 2-80. General system error codes and messages are given in the ND6600 Data Acquisition and Processing Manual, p. 5-198. A list of compiler switches (suppress program listing, read 80 columns instead of 72, etc.) and definitions are given in the manual Program Generation, p. 2-2. The Midas/Fortran compiler uses the following logical units:

Logical UnitFunction

- 2 Source input
- 3 Object output
- 6 Listing output
- 9 Scratch

Midas/Fortran uses logical unit 9 as its overlay input unit. Any user file/element assigned to LU-9 before compilation will not be affected by the compiler program, but it will no longer be assigned to LU-9 when compilation is finished. The Midas/Fortran compiler conforms to the

specifications of the American National Standard Fortran x-3,9-1966, and it is equivalent to the compiler offered by Digital Equipment Corporation to run under their RT-11 operating system. Minor differences from the Fortran rules (DEC PDP-11 system Fortran Language Reference Manual DEC-11-CF-RA-C-D) are discussed in the ND manual Program Generation, p. 2.9.

If there are no error messages, the object code is ready to be saved for linking. Copy it to a file, i.e.,

```
COP SCR2,OBJ.MAIN
```

If you want to save the source code for the main routine, copy it to another file:

```
COP SCR1,SOURCE.MAIN
```

Then, enter the code for the subroutine:

```
R EDITOR
I
SUBROUTINE ADDIT(A,B,C)
C = A + B
RETURN
END
]
EX
```

Compile the subroutine:

```
R FORTRAN
```

```
#CR
```

Copy it to the object file:

```
COP SCR2,OBJ.SUB1
```

To save the source code, type

```
COP SCR1,SOURCE.SUB1
```

6.2.4 Set up a Link File

The linker uses the information it reads from logical unit 8 to link the program. For the sample program, use the editor to create the link file:

```
R EDITOR
```

```
I
```

```
!F (This denotes a Fortran program)
```

```
OBJ.MAIN (list the object codes)
```

```
OBJ.SUB1
```

```
&AUTO.FORLIB (This is the library file for Fortran)
```

```
$
```

```
]
```

```
EX
```

Note that if you use any of the ND system subroutines you should insert the line

```
&AUTO.EXTLIB
```

above the line &AUTO.FORLIB. Now the link file is in SCR1. Create and copy the code into LINK.TEST:

```
COP SCR1,LINK.TEST
```

6.2.5 Link the Object Code to Produce the Executable Code

Make sure that logical unit 8 is defined as LINK.TEST; then, run the linker:

```
RUN APS.NDLINK
```

This program requires a minimum of 2 minutes for execution. The linker printout will appear on the file assigned to logical unit 6. The linker writes the executable code to logical unit 11. Thus, SCR3 will contain this code.

6.2.6 Run the Program

```
RUN SCR3
```

If desired, save the executable code:

COP SCR3,SOURCE.ETEST

If the program does not work properly, re-edit and generate the program.

6.3 READING SPECTRA -- DISK FILE FORMAT

The ND6600 was purchased with eight interfaced ADCs. By pressing the acquire button, data are acquired directly into memory. By using the Nuclear Data subroutine APS.WRITE (see Sect. 4.5), the spectrum is written to disk in a fixed format.

6.3.1 Disk File Format

A spectrum consists of a header and N channels of data. The header is 128 records long; there are 2 words in a record; 2 bytes in a word. There are 256 bytes in a block. Thus, a header contains 2 blocks or 512 bytes. This is true only for MIDAS D+ and earlier versions of the operating system. For MIDAS E+, the header consists of 3 blocks -- an extra block has been added to record efficiency data. Since we do not use these data, we operate under MIDAS D+.

We do not use all the items recorded in the header for MIDAS D+. These items are used by Nuclear Data physics applications programs. Since we have our own applications programs that need items recorded in a header, we have overwritten some of these items. The ND spectrum header file format is shown on Table 1. The items we use are marked with an asterisk. Currently, records 82-116 are overwritten with the header for our data base and a comment by the user.

The program APS.WRITE writes the header with the spectral data.

Some of the header information comes from memory (number of channels in spectra, live time, ADC#, etc.) and the rest comes from the header of the file defined to be LU-8. Thus, we first write the header for the data base and comments to a spectral file defined to be LU-8. Then, when APS.WRITE is run, it copies this information into the file defined to be LU-12. (For additional information, see Sect. 4.5, Recording the Data.) Data types used in the header are defined below:

Code	Name	Definition
F	Real*4	DEC PDP-11 single precision floating point
A	Logical*1	Alphanumeric 1-byte format for character data
I	Integer	Unsigned 16-bit integer
S*4	Spectral	Special ND6600 32-bit integer
U	Unsupported	Not currently used by ND
B	Binary	Bits have individual significance
(One other data type is commonly used in ND programming)		
I*2	Integer*2	DEC PDP-11 1-word integer

The spectral data format is an unsigned 32-bit integer. The two most significant bits are the intensification and marker bits used in the display of the data. The spectral data format (along with a PDP-11 Integer*4 two's complement signed 32-bit word) is shown in Fig. 3.

6.3.2 Reading and Writing Spectra

Before reading or writing spectra, one must define a logical unit number. This may be done external to the program by typing

```
DEF 2 DATA.SPEC
```

To do this in a Fortran program the subroutine DEFY has been written:

```
CALL DEFY(ILU,NAME)
```

ILU is the logical unit and NAME a 16-character array.

To read the spectrum use the subroutine RED:

```
CALL RED(NL,NH,Y)
```

where NL is the lower channel number, NH is the upper, and Y is the floating point array for the data. To write data to a spectral file from a program:

```
CALL WRIT(NL,NH,Y)
```

It is possible to read all the spectra in a file without entering the element name of each. This is done by a subroutine that reads the system directory from SCR2. First place the directory on SCR2:

```
FDI FNAM,,SCR2
```

where FNAM is the name of the file. Then the name of each element may be obtained by using the subroutine DIRK:

```
CALL DIRK(NAM,NUM)
```


This will return the name of the NUMth element. NAM will be in the form FNAM.ELEMENT. NAM is a 16-character Logical*1 array.

7. HARDWARE AND SOFTWARE MAINTENANCE AND TROUBLESHOOTING

7.1 STARTUP, PROBLEMS, AND DIAGNOSES

7.1.1 Startup and Shutdown

To start the computer when it is turned off:

- (1) Turn on the power switch at the bottom rear of the cabinet.
- (2) Turn on the air conditioning switch at the bottom rear of the cabinet. (Do not operate the air conditioning unless the computer is running.)
- (3) Turn on the disk drive.
- (4) When the safe indicator lights, open the unit by grasping the horizontal bar on the front panel and pulling down and out.
- (5) Turn off the write protect switches.
- (6) Insert the disk pack and close the front panel.
- (7) Set the thumbwheel switch setting to 1.
- (8) Depress the RUN/STOP button. When the disk drive reaches operating speed, the READY indicator lights.
- (9) Turn on the power switch on the back of the DAS terminal.
- (10) Then, bring up the system:
 - (a) On the DAS terminal, press down the key labelled break and press 0 at the same time.
 - (b) Still holding the break key, press 1.
 - (c) Press the key labelled autoload.

When the system comes up, it should say:

MIDAS 50K of memory available.

Ideally, the system should never be shut down. The mechanical peripherals should be turned off when they will not be used for a long time, but the electronic parts of the system should be left on at all times. The system should be shut down whenever a printed circuit board is removed or inserted and before any connectors in the system are connected or disconnected.

To power down:

- (1) Transfer all information you wish to save from memory to disk.
- (2) Turn off the disk by pressing the button marked RUN/STOP on the disk drive front panel. Wait until the SAFE indicator light comes on before removing the disk pack.
- (3) Remove the disk pack. If the disk pack is not going to be used for some time, it should be stored on edge to prevent dust from settling on top of the pack.
- (4) Turn off the disk unit.
- (5) Turn off the main power switch at the bottom rear of the cabinet.
- (6) Turn off the air conditioning switch at the bottom rear of the cabinet.

When bringing the system up, pressing break and 0 on the DAS terminal turns on the LSI-11 on the MMI board. Pressing break and 1 turns on the LSI-11 on the DIB board. Striking autoloader causes the firmware autoloader program (firmware located on the MIP board) to bring

a software boot program from disk into memory. The boot program then brings the MIDAS operating system from disk into memory. MIDAS then sets up the system.

The autoloader program always looks for the boot program on the disk designated DK0. The ND6600 we have has two disks — one fixed and one removable. A shorting pin which goes into a socket on the corner of the HDB board is used to select which disk is DK0. If the pin is out, DK0 is the fixed disk and DK1 is the removable disk. If the pin is in, DK0 is the removable disk. Since it is probable that one will lose or break the shorting pin, we attached a cable to the socket with a switch that can be toggled to the IN or OUT position. Normally, it is run in the OUT position, and the fixed disk is DK0 (the system disk) and the removable disk is DK1 (the user disk). The system disk contains both system and user files.

7.1.2 Problems and Diagnoses

A table of system problems and probable causes can be found in the Disk Diagnostic System (DDS) manual on p. 2-14. One problem with a number of causes is that the system will not autoloader. In this section we will examine this problem in detail. If the system will not autoloader, the cause could be one of the following:

- (1) The temperature is too high.
- (2) The system has not settled after installation of a board.
- (3) The software on DK0 is defective.
- (4) The LSI-11 on the DIB or MMI board is defective.
- (5) The disk controller is not working.

- (6) The system voltages are incorrect.
- (7) There is a blown fuse on one of the boards.
- (8) One of the boards is malfunctioning.

First, check the temperature. A thermometer is kept in the front of the air conditioner door. It should read between 50 and 65°F. If the temperature is correct, try turning the system off for about ten minutes and then turning it back on. Wait 10 more minutes. Then, try to autoloading. If this does not work, check the lights in the rear of the system on the MMI and DIB boards. These lights are distributed as follows:

MMI:	RUN light (top of board)	DIB:	RUN light
	MMON light		DMA light
	POLL light (bottom of board)		REF light
			POLL light

After the power is turned on, the two POLL lights should be on. After pressing autoloading, the three lights on the MMI board and the four lights on the DIB board should be on. These lights are on the edges of the board nearest you. There is an interrupt light at the far edge of the MMI board that should not be on. If it is, turn off the system, wait, check the temperature, and reload. If the interrupt light is not on, you will probably not be able to see it without removing the board.

The top lights on both the MMI and DIB boards are the run lights. If the MMI light turns on momentarily when AUTO is pressed, the autoloading command is communicating, but MIDAS is not loading or will not run. This could be due to defective software on DKO or to a bad memory

board. Check the software by installing the backup disk copy of the system and changing the shorting pin to IN so DK0 is now the removable disk. Try to autoload from this system disk. If this does not work, it is necessary to run further checks. Since it is better to avoid removing and installing the boards, the checks involving these should be done last. Thus, we next check the system voltages, use the Disk Diagnostic System to see if the LSI-11s are running properly, check the memory boards, and then check for blown fuses on the boards.

There are test points for the supply voltages at the DKI board. These voltages and their values are:

-12VDC (top test point)

+12VDC

+ 5VDC

-15VDC

+15VDC

+ 5VDC

GND (bottom test point)

In addition, there is an auxiliary power supply at the left rear door of the cabinet that should read +5VDC.

7.2 DISK DIAGNOSTIC SYSTEM

The DDS can be useful in locating a defective board, disk drive, or other piece of hardware. The DDS, a disk-based monitor system, is controlled by English-like commands and communicates with the user via a nonbuffered, noninterrupt-driven Input/Output. This section will discuss the ways DDS can be used, recommended routine diagnostic tests,

and the memory map of our current configuration. In addition to its diagnostic testing functions described below, DDS can be used to examine memory locations, to deposit instructions in memory, and to execute manually a program (see DDS manual, pp. 2.3 and A-1). Simple LSI-11 run tests (shown below) can be performed with DDS by loading instructions and observing the lights on the back of the MMI and DIB boards.

The DDS has a series of test routines for the LSI-11s, the memory boards, and the disks. There are three ways to use these test routines: (1) run individual tests, (2) use LINK files which run several tests, and (3) use USE files which are similar to job streams or command files.

Note that when one has a problem of unknown hardware origin it is recommended to perform DDS system checkout in the following order: LSI-11, memory, and then the disks. The LINK files discussed below provide the most convenient way to do this.

7.2.1 Installation of DDS

Before running any diagnostic tests, first install DDS using either the DAS terminal or the DECwriter. The following steps are necessary:

- (1) Press BREAK and 0 simultaneously on the DAS keyboard; then, press BREAK and 1 simultaneously.
- (2) Remove the upper platter.
- (3) Insert the DDS disk.
- (4) Switch the shorting pin to IN.

- (5) Press the key labelled AUTO on the DAS terminal to load DDS.
- (6) Press the SPACE bar and then the REPEAT key on the DAS terminal.

If you wish to have a hard copy, you can run DDS from the DECwriter.
To do this, substitute the following for step 6:

Press the space bar on the DECwriter repeatedly.

The line

DDS V02-02 17-OCT-1980

should appear. If you are using the DECwriter you should then type

MMON

On the DAS terminal this is done automatically. This turns on memory management. To turn it off, type

MMOFF.

7.2.2 Simple LSI-11 Tests

To check the MMI LSI-11:

Type, at the same time,

control, HERE IS, 0

(0 indicates that you wish to communicate with the LSI-11 on the MMI board.)

Type, while watching the MMI run light flash,

0/0cr 0g

(This instruction executes a halt.)

Then type

0/777cr 0g

The MMI run light should remain on. (This instruction executes an infinite loop.)

Then type, together

Break 0

The MMI light should turn off.

To check the LSI-11 on the DIB:

Perform the same operations as above, except for two lines:

control, HERE IS, 0 becomes control C, HERE IS, 1

and at the end

Break 0 becomes Break 1

Watch the run light on the DIB board while executing these instructions.

7.2.3 Individual DDS Tests

To run a single diagnostic test, type RUN MM-NNNN, where MM-NNNN is the number of the diagnostic you wish to run. For example, to read and verify the disks, type

RUN 44-0151

The system will perform this test and respond:

READ / VERIFY ON DK0 - 100 TPI DISK TYPE = DDS

READ / VERIFY ON DK1 - 100 TPI - DISK TYPE - MIDAS SYS

44-0151 COMPLETED

if there are no errors. If there are errors, they will be listed. The diagnostic tests and their numbers are listed in the DDS manual.

7.2.4 LINK Files

To run a LINK file (DDS manual, p. 2.6) type RUN and the name of the file. For example, the LINK file LSI-11 runs all the LSI-11 tests. Type

RUN LSI-11

The system will respond:

LSI-11 basic instruction test run time = 60 sec. 44-0720
completed

It will then run and indicate completion and possible errors for four more LSI-11 tests: the extended instructions, add/subtract, multiply, and divide. Note that LINK files are just lists of tests which are run consecutively.

7.2.5 USE Files

USE files are similar to job streams in that they contain commands as well as tests which are run. A set of USE files comes with the DDS package and the programmers may create more.

To list the USE files type

USE 2

To see the contents of a particular file, type SEE and the number of the file. As an example, the file USE 35 will be examined and discussed. To see USE file 35, type

SEE 35

The system will reply:

```
35  Msg'**** test w/r 0 or 177777 pattern to memory ****'
    Msg'**** v = board v2 = pattern v2 = 1 pattern = 0 ***
    *'msg'**** else pattern = 177777 ****' 1 44-0159(v) b
```

```
060000 ifv2=1 d 0464 1 d 0604 1 go exit else d 0464 2
d 0604 2 go
```

All the DDS commands are defined in Appendix A of the DDS manual.

To run USE 35 type

USE 35

The system will reply:

```
**** test w/r 0 or 177777 pattern to memory ****
**** v = board v2 = pattern v2 = 1 pattern = 0 ****
**** else pattern = 177777 ****
ready
Memory 4k segment run time - 280 seconds
44-0159 completed
```

This particular test wrote and read 0s in all of memory except the section where DDS resides. In this case v = 0 and v2 = 1 since these values are set when the system autoloads. To change v2 to 2, type

V2=2

Then, the pattern 177777 will be written and read in memory when USE 35 is run. The DDS loads test 44-0159, uses octal 060000 as a base when depositing information; if v2 = 1, it deposits 1 in locations 0464 and 0604, runs the test, and exits. If v2 is not equal to 1, it deposits 2

in locations 0464 and 0604 and runs the test, reading and writing pattern 177777.

7.2.6 Routine Diagnostic Checkout

The LINK files should be run routinely to check for possible problems. The LINK files (DDS manual, p. 2.6) are: LSI-11, which runs for 5 minutes; MEMORY, which runs all the memory tests that take seconds and minutes to run; and DISKP, which sees if the disk and disk controller can read and position correctly. To run these LINK files, type RUN and the name of the file; wait for DDS to indicate completion of the test; then, run the next test, i.e., type

RUN LSI-11

RUN MEMORY

RUN DISKP

7.2.7 Memory Configuration

To get a memory map, type

RUN 44-0158

The system will reply:

system memory map n/a=not available o/l=on line

segment	sw1	sw2	sw3	sw4	sw5	sw6	sw7	sw8
4k	o/l	o/l	o/l	o/l	o/l	n/a	n/a	o/l
8k	o/l	o/l	o/l	o/l	o/l	n/a	n/a	n/a
12k	o/l	o/l	o/l	o/l	o/l	n/a	n/a	n/a
16k	o/l	o/l	o/l	o/l	o/l	n/a	n/a	n/a

44-0158 completed

In our system we have 80K words of memory. Four 16K word segments are on one board (sw1-sw4) and one is on the other (sw5). At the address of sw8 is the 4K of internal memory on the DIB LSI-11 that is used to manage the display and acquisition of data. There is also 4K of internal memory on the MMI LSI-11 that is not used. If an error is suspected in the internal memory of the DIB it can be exchanged with the memory on the MMI. Note that DDS resides in the 16K of sw1. To test this portion of the memory, remove the memory board (turn off the power first) and change the toggle switches so that 1 reads 4, 2 reads 3, 3 reads 2, and 4 reads 1.

When a malfunctioning board is located, it can be exchanged for a new one from Nuclear Data. If problems with the disk are encountered, further tests can be run independently of the ND6600 system. This is done by using a disk exerciser.

7.3 COPYING AND INITIALIZING DISKS

To copy or duplicate from the fixed disk (DK0) to the removable disk (DK1):

- (1) Write protect the lower disk (DK0).

(2) Put the new disk in DK1.

(3) Enter DUP_{er}

Respond

DK0,DK1,f

(where the first argument, DK0, is the source disk from which data is to be copied).

The second argument is the destination disk.

f indicates the destination disk is to be formatted.

When complete, MIDAS will ask for arguments for another duplication. Respond with a carriage return to exit.

(4) Toggle offline and back on to read in a new directory.

(MIDAS stores the directory in memory and refers to this directory rather than rereading it from disk each time it accesses the disk.)

To make a copy of a removable disk on another removable disk, copy from the original removable disk to the fixed disk and then from the fixed disk to the new removable disk. Since the current system is on the fixed disk, first copy it to a spare removable disk and then copy it back after duplicating the desired disk. The following is a blow-by-blow account of the actions required in accomplishing this objective:

(1) Copy the system disk from the fixed disk to a removable disk

(Steps 1-4 above).

(2) Insert the disk to be copied.

(3) Enter DUP.

(4) Respond DK1,DK0

- (5) System will say: Press carriage return when ready. Press CR.
(This duplicates from the removable to the fixed disk.)
- (6) When the system says Enter arguments, remove the original disk.
- (7) Put in the new user disk.
- (8) Respond DK0,DK1,f
- (9) Press CR. (This formats the user disk and copies from the fixed to the removable disk.)
- (10) When system says Enter arguments, remove new user disk.
- (11) Put in copy of system disk.
- (12) Now respond DK1,DK0.
- (13) When ready, press CR. (This restores the system to the fixed disk.)
- (14) When system says Enter arguments, press CR.
- (15) Remove DK1 and replace with a user disk.
- (16) Toggle DK1 off and online so the directory of DK1 can be read.

For further information, see ND6600 Data Acquisition and Processing System, pp. 5-15, 5-76, and 5-112.

7.4 DISK MAINTENANCE

The disks should be checked in the following manner:

- (1) Bring the disk drive to the SAFE condition, remove the cartridge, and turn off all power. This includes power to the mainframe. The most convenient method for doing this is using the main power switch at the rear of the cabinet.
- (2) Pull the disk drive out on its slides until it is clear of the

cabinet. It is not necessary to disconnect the ribbon cable, but care should be taken to ensure that the cable is not damaged as the disk moves out. The drive should clear the cabinet by no more than two inches. If your system has more than one disk, DO NOT PULL BOTH DRIVES OUT AT THE SAME TIME. This would unbalance the system and could result in the cabinet tipping. Once the drive is out on the slides, remove the metal bonnet by removing seven Phillips screws holding the bonnet onto the unit. Loosen the two slotted screws, release the spring lock, and raise the main logic board to a vertical position.

- (3) Use the inspection mirror and pen light provided in the preventative maintenance kit for checking the condition of the heads.
- (4) If, upon inspection, all heads are free of oxide build-up, leave them alone. Heads should be cleaned only as needed.
- (5) Use only isopropyl alcohol (minimum 91% by volume) and bonded cotton swabs, of the type provided in the maintenance kit. DO NOT use the normal q-tips that can be purchased at any drug store. There is a danger that loose fibers may contaminate the heads or platters.
- (6) With a swab soaked in alcohol, rub the head in a circular motion to clean any oxide build-up. To ensure that the holes in each head are cleaned, use a short circular motion of the swab into the hole. Use a dry swab to buff the head.
- (7) Inspect the cleaned heads to ensure that no lint or other contaminant was accidentally left on or around the heads.
- (8) Remember: DO NOT touch the heads with your fingers.

- (9) Remove the inspection screen toward the left side of the drive. Slowly turn the lower or fixed platter by turning the center hub. Check the lower platter for scratches, continuous circle burn marks, and head hits in the data and loading area. If there are more than six hits, the platter should be replaced.
- WARNING: Do not blow on the platter to remove lint particles. Moisture from your breath could cause a hazardous condition.
- (10) Before reinstalling the inspection screen, check it for any loose contaminants.
- (11) Inspect the prefilter. This is done by removing the front cover of the disk drive. This requires the loosening of six allen screws, three on each side. Then, gently remove the cover using a screwdriver to hold in the door lock bolts.
- (12) If the prefilter needs cleaning, remove it and clean with a vacuum cleaner. If this does not remove all of the dust and dirt, it can be washed in clean water. DO NOT use any cleaning agents; soaps, cleaning fluids, etc. IMPORTANT - let the prefilter completely air dry; never install it while it is still damp.
- (13) The absolute filter cannot be cleaned. In an average environment, it should be changed every six months. In a dirty environment, it should be changed frequently. If the back of the absolute filter begins to look gray, it has long since lost its effectiveness. Replace it, and reduce the time period between replacements.
- (14) Check and clean the area where the absolute filter resides, using pads or swabs with isopropyl alcohol.
- (15) Remember to tighten all screws and check to ensure none of the connectors are loose.

- (16) Release the spring-loaded latch holding the logic board, lower it into place, and tighten the slotted screws. Reassemble the drive and gently restore it to the cabinet, taking care not to pinch the ribbon cables.
- (17) Turn on the power.

7.5 OPERATION AND MAINTENANCE OF THE AIR CONDITIONER

The air conditioner used by the ND6600 is a 7000-Btu Loboy(R) air conditioner designed for electronic enclosures. It is manufactured by McLean Engineering Midwest. The air inside the enclosure is cooled and returned to the enclosure. Thus, this air should be cleaner as well as cooler than ambient air. In addition to this air, ambient air is drawn through a filter, across the condenser coils, and discharged back into the environment.

There are five items which need to be noted for effective use of this unit:

- (1) This unit is not to be operated without a heat load.
- (2) After the unit is turned off, wait 1 minute before turning it on again. This will allow the refrigeration system's internal pressures time to equalize and will avoid compressor damage.
- (3) The front filter should be cleaned or replaced periodically (every six months in our EBT-S environment).
- (4) The system has a temperature monitor-alarm circuit. There is a temperature dial which can be set to the maximum desirable temperature inside the enclosure (95°F). If this temperature is exceeded, a siren will sound. To reset, turn off power to the ND6600 and turn it back on. Then check the thermometer in front

of the unit. If this reads less than 65°F, the siren may have been a false alarm. This could be due to a transient drop in voltage or noise on the line.

- (5) There is an adjusting screw under a red plastic cap in front of the air conditioner behind the filter. This must be set correctly for the unit to operate properly. If the unit runs too hot or too cold and if the freon and filter are not at fault, this may need adjusting. First, obtain the factory setting for this hot gas bypass valve. Second, turn the screw all the way clockwise. Third, back off one full turn. Fourth, wait 10-30 minutes and check the temperature. If it needs to be cooler, back off more. Do not back off more than two full turns.

For further information, refer to the instruction manual or contact McLean Engineering Midwest, 9560 85th Avenue North, Maple Grove, Minnesota 55369, phone 612/425-4747.

Appendix: COMMAND SUMMARY

In this report, system commands have been explained and examples have been given for specific applications. In this appendix, the general form of these commands will be given. For more detailed information and other commands, see the Nuclear Data manuals. Note that the arguments in parentheses are optional.

AADC: allows a user to assign an ADC and ADC memory to hold spectral data.

AADC AD (,GP,CH,BI,PL,PR)

AD System number of the ADC assigned.

GP Number of memory groups for this ADC (1 through 8192).

CH Number of channels per group (1 through 8192).

BI Number of bits per channel (1 through 30).

PL Preset live time in seconds.

PR Preset real time in seconds.

ADON: Turns an ADC/Group on.

ADON (A1/G1, A2/G1, A5/G3)

A1 System ADC number.

G1 Group number associated with A1.

AOFF: Turns ADCs off.

AOFF (A1,A2, A8)

A1, A2, ... System ADC numbers

ASSIGN: Creates a valid user number and assigns a priority limit.

ASSIGN UN, PR (,PW, SF, FPW, SPW)

UN User number to be created (0-249)

PR New user's limit priority (1-15).

PW New user's password (1-6 alphanumeric characters; first character must be alpha).

SF New user's auxiliary command table.

FPW Password for the auxiliary command table.

SPW Special password (for user 0 only).

BYE: Removes this console from control of the specified user and frees it for use by another user.

BYE UN

UN The number of the user currently signed on this terminal.

COPY: Copies data from a file or element on a disk to another file or element on a disk.

COPY SF(,DEV),DF(,DEV)

SF Source file.

DEV Disk unit in which the preceding file (SF or DF) resides.

DF Destination file.

DEFINE: Assigns a physical file to a logical unit number.

DEFINE LU,SF(,DEV,PW)

LU Logical unit number (1-12).

SF File name and optional element name to be equated to the logical unit.

DEV Disk unit on which the file resides.

PW Password of the file or element.

DELETE: Deletes the directory entry of a disk file or element.

DELETE SF (,DEV) to delete an element.

DELETE SF*(DEV) to delete a file.

SF File or element.

DEV Disk unit on which the file resides.

DIRECT: Prints some of the attributes of a disk unit, a file, and/or
the elements of a file.

DIRECT (SF,DEV,LST,LS)

SF Name of the file or element.

DEV Disk unit on which SF resides, or if SF is not specified, DEV is
the disk unit to be directoried.

LST Name of the output device.

LS List switch: S = just files, A = files and elements.

FDIRECT: Displays a directory of all the attributes of a disk unit, a file
and/or the elements of a file.

FDIRECT (SF,DEV,LST,LS)

SF File or element.

DEV Disk unit on which SF resides.

LST Name of the output device where the directory is printed.

LS List switch: S = just files, A = files and elements.

FILE: Creates a file on a disk.

FILE SF,DEV,SS,LN(,CC,PT,PW,DT,UN,RS)

SF Name of the file to be created.

DEV Disk unit where the file is to be located.

SS Starting sector of the file (0-9743 for hard disk).

LN Length of the file in sectors. Min = 8 for directoried files.
 Max = 9734 for hard disk.

CC File type: D = Directoried (default). ND = Nondirectoried.

PT Protection: A = Absolute. R = Read protect. W = Write protect.
 E = Execute protect. Default = none.

PW Password (1-6 alphanumeric). Default = none.

DT Data type (0-255). Default = 10, undefined.

UN Owner of the file (0-249 or 255). Default = requesting user.

RS Record size (0-255). Default = 0.

HELLO: Signs a user onto the system.

HELLO UN (,A)

UN Valid user number. (0-249)

A Autorun specification (if specified, it must be A).

LUPRINT: Displays the current logical unit assignments.

LUPRINT (SRT,NBR)

SRT Starting logical unit number to be displayed.

NBR Number of logical unit numbers to be displayed.

PACK: Eliminates any deleted elements within a file.

PACK DF(,DEV)

DF Directoried file.

DEV Disk unit on which file resides.

WHO: Determines which user is signed on a terminal.

WHO

TABLE 1

LU-8/LU-12: ND6600 Spectrum Header File Format

From Record	To Word	Record	Word	Data Type	Content
1	1	1	1	I	*Must be zeroed
1	2	2	1	I	*System Identifier
2	2	2	1	I	*Subsystem Identifier
2	2	6	2	U	Reserved for tape compatibility
7	1	7	1	I	*Byte No. Offset where spectral data begin
7	2	7	2	I	*User No.
8	1	8	2	S	*ADC Number (Bit Mask)
9	1	9	1	I	Configuration No.
9	2	9	2	I	*No. of bytes/channel
10	1	10	1	I	*No. of channels/group
10	2	10	2	I	*No. of groups/configuration
11	1	11	1	I	Digital Offset
11	2	11	2	B	*Preset Status
12	1	12	2	S	*Preset Live Time in seconds
13	1	13	2	S	*Elapsed Live Time in seconds
14	1	14	2	S	*Preset Real Time in seconds
15	1	15	2	S	*Elapsed Real Time in seconds
16	1	16	2	S	*Preset Total
17	1	17	2	S	*Actual total counts in region
18	1	18	2	S	*Preset Level
19	1	19	2	S	Actual count level of peak in region
20	1	10	1	I	*Preset total/level start channel
20	2	20	2	I	*Preset total/level end channel
21	1	21	1	I	*Acquisition Start Date (Julian 1-366)
21	2	21	2	I	*Acquisition Start Year
22	1	22	2	S	*Acquisition Start Time of Day (Seconds after midnight)
23	1	27	1		Reserved for ADC parameters
27	2	27	2	I	Application Code (Note 3)
28	1	28	2	F	Energy Calibration offset (keV) (Note 4)
29	1	29	2	F	Energy Calibration Slope (keV/channel) (Note 4)
30	1	30	2	F	Quadratic coefficient of energy vs channel curve (keV/channel ²) (Note 4)
31	1	31	2		Reserved for additional energy calibration parameter
32	1	32	1		Not used
32	2	36	1		Reserved for ND600 and ND660 Systems
36	2	44	2		Not used
45	1	45	2	A	Unit of Activity
46	1	46	2	F	Activity Multiplication Factor from μCi to Current Unit

LU-8/LU-12: ND6600 Spectrum Header File Format (Cont'd)

From Record		To Word	Record	Word	Data Type Content
47	1	47	2	F	Pulse Pileup Slope Factor
48	1	48	2	A	Fit singlets with Gaussian ("YES" or "NO")
49	1	64	2		Special Use
65	1	65	2	F	Sensitivity
66	1	66	2	F	Gaussian Sensitivity Parameter in %
67	1	67	2	F	Number of Iterations
68	1	68	2	F	Peak Search Start Channel
69	1	69	2	F	Peak Search End Channel
70	1	70	2	F	Half Life Ratio
71	1	71	2	F	Abundance Limit (%)
72	1	72	2	F	Energy Tolerance (keV)
73	1	73	2	F	Sample Quantity
74	1	74	1	I	Sample Date (Julian 1-366)
75	1	75	2	S	Sample Date Time of Day (seconds after midnight)
76	1	76	1	I	Calibration Date (Julian 1-366)
77	1	77	2	S	Calibration Date Time of Day (seconds after midnight)
78	1	78	2	F	FWHM Offset (Note 5)
79	1	79	2	F	FWHM slope (Note 5)
80	1	81	2		Reserved for FWHM parameters
82	1	83	2		Unused
84	1	99	2	A	Sample Title (63 characters)
100	1	103	2	A	Sample ID (15 characters)
104	1	107	2	A	Sample Type (15 characters)
108	1	111	2	A	Sample Units (15 characters)
112	1	115	2	A	Sample Geometry (15 characters)
116	1	119	2	A	Efficiency File Name (15 characters)
120	1	123	2	A	Detector ID (15 characters)
124	1	127	2	A	Library File Name (15 characters)
128	1	128	2		Unused